Marcel Richter

M.Sc. Computer Science Student | Embedded Software & Security

marcel.ric@gmx.de +4915170686581 Munich Available now, up to 16 h/week

linkedin.com/in/marcel-ric github.com/Ma161ric

PROFILE

Computer science Master's student (LMU Munich) focusing on embedded systems, parallel systems and security. 1.5 years of industry experience in firmware development for NAND flash controllers at Hyperstone. Low-level programming in C/C++ and assembly, microcontrollers, FPGA/VHDL, as well as communication and bus systems.

WORK EXPERIENCE

Working Student, Software/Firmware, *Hyperstone GmbH* 03/2023 – 02/2024
Konstanz

- Simulated controller firmware to analyse and optimise RAID performance
- Implemented and tested various RAID configurations for NAND flash storage
- Developed a Python model to evaluate firmware behaviour

Intern (Practical Semester), *Hyperstone GmbH* 09/2022 – 02/2023
Konstanz

- Cross-compiled and tested firmware tools (hsfmt)
- Built up knowledge of NAND flash storage architectures and RAID concepts

EDUCATION

M.Sc. Computer Science, *LMU Munich* 04/2026 – 09/2028 (expected)
Focus on systems & security, incl. Program Analysis for Security, Verification of Parallel Programs

B.Sc. Applied Computer Science, specialisation in Embedded Systems, *HTWG Konstanz* 10/2020 – 02/2026
Tutor for Digital Logic (winter semester 2024/25)

PROJECTS

Embedded lab projects, *HTWG Konstanz, Embedded Systems specialisation*

- Low-level programming of microcontrollers (MSP430) in C and assembly
- Interfacing peripherals via UART, SPI and I²C
- Embedded Linux on the Raspberry Pi; circuit design in VHDL for FPGAs

Bachelor's thesis: ML pipeline for rating chat messages, *HTWG Konstanz* 02/2026

- Leakage-free Python pipeline (scikit-learn, Sentence Transformers) for ~2,500 messages across 20 criteria
- Synthetic training data from a local LLM (Gemma 3 via Ollama); evaluated four models with 6-fold cross-validation

flowculate – web app for customer path analysis, *Team project, HTWG Konstanz, grade 1.0* 04/2024 – 09/2024

- Developed in a team of five; built the frontend with the survey flow and layout editor in Next.js and TypeScript

SKILLS

Programming
C, C++, Python, Assembly, VHDL; also Java, Scala, TypeScript

Embedded
Fundamentals: microcontrollers (MSP430), Embedded Linux (Raspberry Pi), UART/SPI/I²C, FPGA/VHDL

Security
Cryptography fundamentals (AES, RSA, TLS), secure coding, fuzzing and pentesting (basics)

Tools
Linux, gcc, gdb, make, Git

LANGUAGES

German (native) | **English** (C1) | **French** (basic) | **Spanish** (A1)